

Solar Desalination Energy Balance

Bench-scale solar still SDU-3

Abstract

An hourly energy balance was assembled for solar still SDU-3 to separate absorbed irradiance, stored heat, distillate production, and Q_{loss} . Thermal efficiency and specific water yield are reported without extrapolation beyond the measured day.

1. Balance definition

Irradiance G_t entered through the transparent cover. Water temperature and condensate mass m_d were recorded every 10 min, while ambient exchange was represented by the residual term Q_{loss} [1].

$$\eta_{\text{th}} = m_d h_{\text{fg}} / (A \int G_t dt)$$

2. Hourly observations

Hour	G _t	T _w	m _d	Q _{loss}
09:00	410	34.2	0.08	52
10:00	612	41.8	0.17	68
11:00	780	49.6	0.29	83
12:00	845	55.1	0.38	96
13:00	802	57.0	0.41	101

Table 1. Observations for SDU-3. G_t is in W m⁻²; m_d is in kg m⁻² h⁻¹.

2.1 Interpretation boundary

The noon efficiency increase coincided with stored heat release. It should not be attributed solely to cover transmission because wind speed also declined.

3. Energy-flow diagram



Figure 1. Energy-flow terms and arrow directions used in the hourly balance. Labels name measured or residual terms.

Footnote 1. Arrow width is schematic and does not encode magnitude.

4. Conclusions

SDU-3 reached its largest specific water yield after peak irradiance, consistent with temporary heat storage. The energy balance closes within 6.2 %, but a single day cannot define seasonal thermal efficiency.

References

- [1] Rahman A. Hourly accounting in basin solar stills. Desalination Methods 17:44-53.
- [2] Saleh M. Direction-preserving energy diagrams. Water Energy Notes 6:73-81.
- [3] Khan R. Reporting residual heat terms. Thermal Practice 10:15-22.